

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
Second Semester 2025-26, OPEN BOOK
EEE/CS/INSTR/ECE F241 Microprocessors & Interfacing
Design Test (Time: 90 minutes)
ID Number:

Name:

Note: Marking for each subsection is given below. Further partial marking within the subsection won't be there. Hence do write the answer very carefully. Ensure completeness of each section to obtain marks. VCC/GND signals need not be shown for chips. Rest of the signals are to be shown.

a) Show the address mapping for all the memory chips clearly (show the starting address and ending address ..name them RAM 1 (including odd even bank) , ROM1 (Including odd even bank) , ROM2 (including odd even bank etc).

RAM1 [1M]: 02 000H - 03 FFFH
 ROM1 [1M]: 00 000H - 01 FFFH
 ROM2 [1M]: FE 000H - FF FFFH

b) Show the address mapping (mention addresses assigned to the I/O chips clearly.)

Mention the addresses assigned to 8255 [1M]

F0H
 F2H
 F4H
 F6H

Mention the addresses assigned to 8254 [1M]

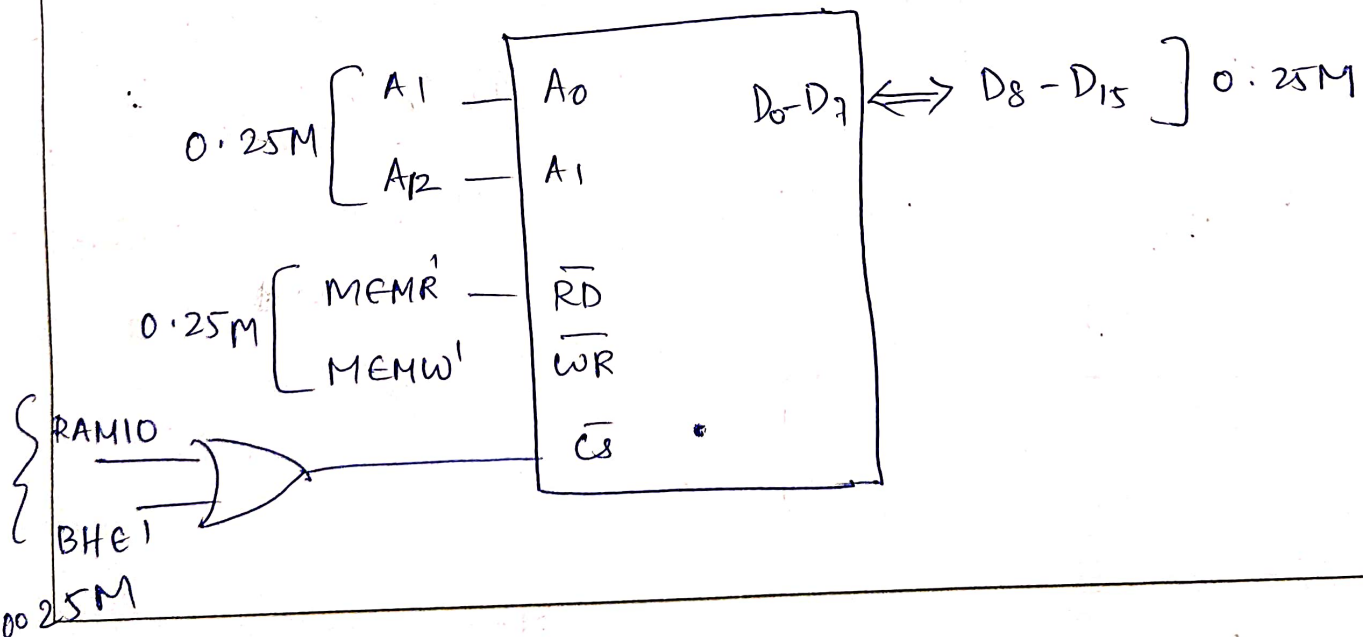
A1H
 A3H
 A5H
 A7H

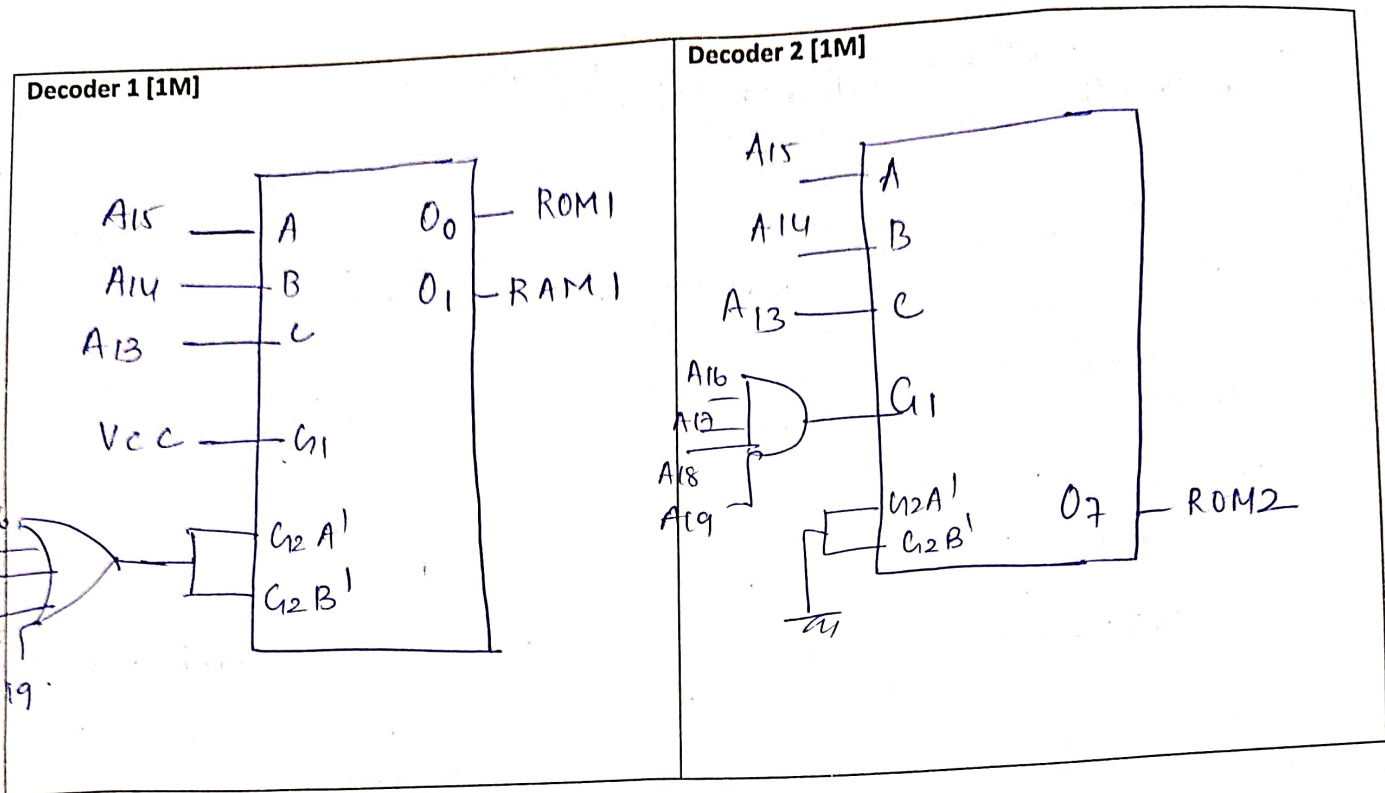
Mention the addresses assigned to 8259 [1M]

A0H
 A2H

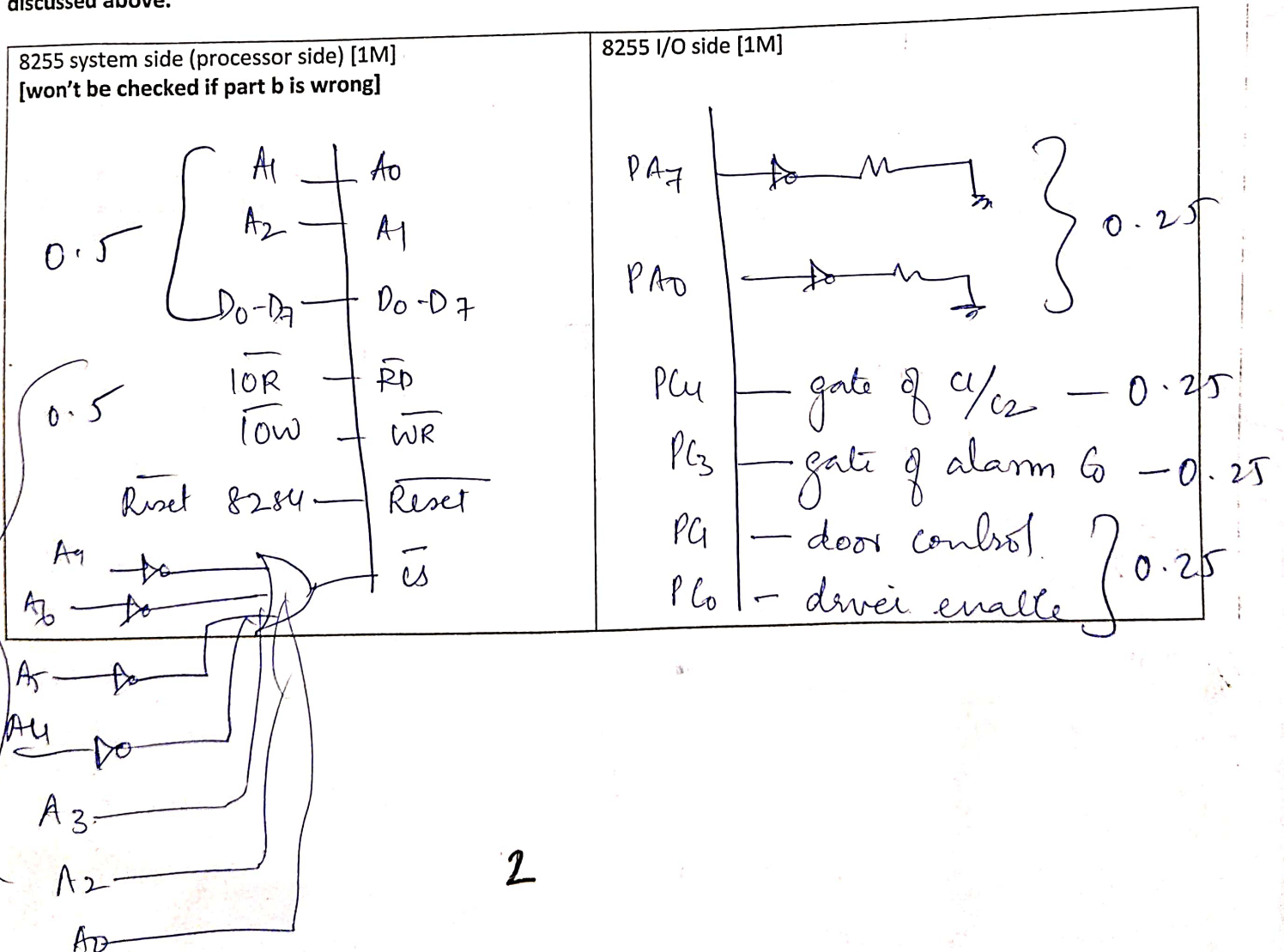
c) Draw a neat memory interfacing logic showing all the required signals. If the system uses LS138, show all the inputs and outputs of the LS138. Show the interfacing of one even and odd bank ROM and RAM with the LS138 and 8086 mentioned below. [won't be checked if part a is wrong]

RAM1 odd memory bank [1M]





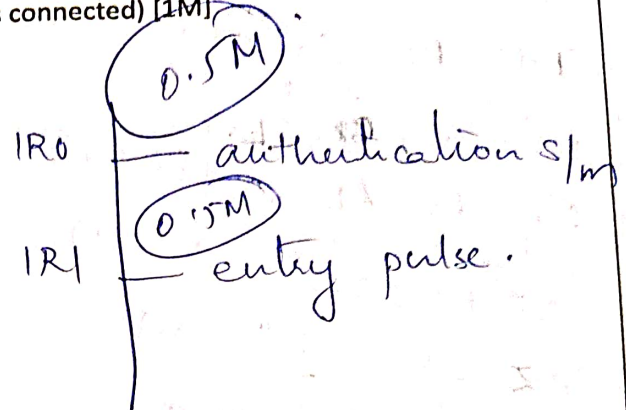
d) Draw the hardware interfacing for the I/O peripherals (both the system side and I/O side) discussed above.



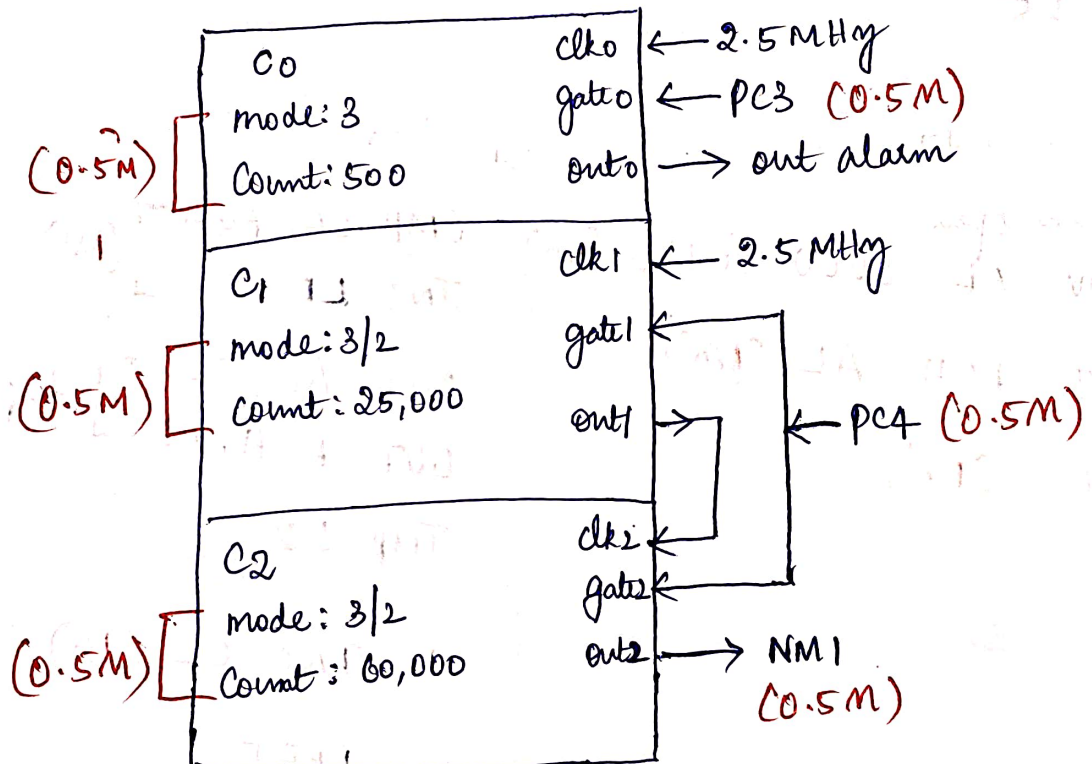
Write the ALP code snippet for the initialization of 8255 [2M]

```
mov al, 10000000b — ①
out F6, al — ①
```

Draw the 8259 I/O side (show where are the INTR Interrupts connected) [1M]



Draw the required 8254 connections (Mention the counter, counter mode, count clock, gate, out connections only) [3M].



Calculation of count:

C0:

$$\frac{1/p}{N} = 0/p$$

$$= \frac{2.5 \times 10^6}{5000}$$

$$= 500$$

C1:

$$\frac{2.5 \times 10^6}{2.5 \times 10^3}$$

$$= 100 \text{ Hz}$$

C2:

$$0.01 \times 60,000$$

$$= 600 \text{ seconds}$$

$$= 10 \text{ min.}$$

ISR in ALP for NMI [2M]

```
INC DL CL CL
; display count
MOV AL, CL CL ] (0.5M)
OUT F0, AL
; Check for 30 min (3 x 10 min)
CMP DL, 3 ] (0.5M)
JNZ L3
; open door and buzzer
MOV AL, 0000 0111b ] (0.5M)
OUT F4H, AL
JMP L5
```

```
; Close door and no alarm
L3: MOV AL, 0000 1001b ] (0.5M)
OUT F4H, AL
L5: IRET
```

ISR in ALP for INTR Vector no 30H [3M]

```
; door open
MOV AL, 0000 1011b (1M)
OUT F4H, AL (1M)
IRET. (1M)
```

ISR IN ALP for INTR (Vector no 31H [3M])

```
CMP CL, 04H ] (1M)
JNZ L1
MOV AL, CL ] display count (1M)
OUT F0H, AL ]
JMP L2
L1: INC CL (1M)
L2: IRET
```

0.5M if CL is increased without comparison